Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2404)

Key

SECTION – I

1. D	7. C	13. C	19. A
2. C	8. B	14. D	20. B
3. B	9. B	15. B	21. B
4. D	10. D	16. A	22. D
5. A	11. C	17. A	23. 4
6. D	12. C	18. C	24. 5

SECTION – II

1. C	6. 2	11. A	16. B
2. 59	7. 4	12. B	17. D
3. 69	8. 3	13. D	18. C
4. A	9. 6	14. B	19. C
5. A	10. 6	15. C	20. A

SECTION – III

1. 9	7. C	13. C	19. B
2. 28	8. B	14. A	20. 32
3. 108	9. C	15. 4	21. C
4. C	10. 35	16. A	22. B
5. D	11. B	17. 1814	
6. 88	12. 14	18. D	

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 532

1. Option A: Businesses should create an experience that is specific and that will come to define the company. However, the author doesn't limit this to just absorbing and immersive. There are also other aspects like the level of participation. Hence, Option A is not the answer.
Option B: The author says that a business must offer an experience that will come to define them. The author doesn't say that such an experience must be 'unique.' It should be 'specific' and clear. Also, the author doesn't touch upon profitability. Option B is not the answer.
Option C: The question of price and what the customer pays has not been mentioned in the last para of the passage. It has been mentioned only in the first para. The author's implication in the last para is about what defines a business, and it is not about how much is the customer's money worth. Option C is not the answer.
Option D: This is what the author was suggesting, that every manager should think about the kind of experience a business can offer, and that ***experience defines the business.*** So, it can be said that businesses must offer a specific experience to their customers that are clear. Option D is the answer. Choice (D)

2. Option A: This is similar to the activity of descending the Grand Canyon. It offers education as well as entertainment. It involves both active participation and immersion in the experience of rafting itself. The author will approve of this categorisation. Option A is not the answer.
Option B: Attending a painting class, is more education than entertainment, and hence more active participation than immersion. The author will approve of this categorisation. Option B is not the answer.
Option C: Attending a yoga class involves more active participation than immersion, and hence it is neither an escapist event nor an aesthetic event.

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It is where students are still outside the event to some extent. Option C is the answer.
Option D: Here, the individual is actively participating and is also immersed in the event – both education and entertainment. This is definitely an escapist event. Option D is not the answer. Choice (C)

3. We are looking for an option that says that audiences at a symphony do contribute to the experience of other audience members. That way symphony and skiing become similar experiences. The author's assumption was that symphonies don't have active participation and, hence, they offer a different experience compared to skiing where audience presence matters as understood from '***simply by being there, they contribute to the visual and aural event that others experience.***'
Option A: While this says that the symphony experience depends on other parameters apart from the musician, we don't know what those parameters are. They could be acoustics, they could be the venue, the ticket price, etc. We are looking for an option that gives importance to the audience. Option A is not the answer.
Option B: The audience at the symphony is also important for the experience according to this option. If true, this weakens the author's conclusion that a symphony presents an example of passive participation. If the audience is important and it influences the musician, then they are not passive anymore, they become active participants. Hence, Option B is the answer.
Option C: This option shows that the audience size doesn't matter as far as the symphony experience is concerned. So, it doesn't weaken the author's conclusion. Option C is not the answer.
Option D: A symphony experience depends on each individual's grasp of music. In other words, here, the presence of other audience members doesn't really affect an individual's experience. Their own understanding of music is what affects their experience. So, option D is not the answer. Choice (B)

4. Option A: Watching a cricket match at a sports bar is more immersing because of the sights, sound, and smells around an individual, something that may not be the case at home. This is the same as the author saying 'seeing a film at the theater with an audience, large screen, and stereophonic sound is more immersing than watching the same film on video at home.' Hence, Option A is not the answer.
Option B: This is similar to the author saying 'Furiously scribbling notes while listening to a physics lecture is more absorbing than reading a textbook.' Being in the class is more absorbing than listening to a video recording of the same lecture since in the first case, a person is physically present and engrossed in the event, and less prone to distractions probably. Option B is not the answer.
Option C: This is very similar to watching a movie at a theatre as compared to watching the same movie at home, without the environment. This option is compatible with the author's examples. Option C is not the answer.
Option D: Attending a football match with friends and attending a football match alone involves the same cues associated with senses, and the same environment. We don't have enough information to differentiate between an experience with friends and without them. This option is not compatible with the author's definitions of absorption and immersion. Option D is the answer. Choice (D)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 518

5. Option A: This is the purpose of the example, which shows that cognitive skills are non-translatable. This has been established in psychology. So, cognitive skills cannot be transferred across domains. Being good at remembering long strings of digits for example cannot be transferred to remembering long strings of letters. Hence, Option A is the answer.
Option B: Domain-specific knowledge has not been the focus of this example. The given example is about cognitive skills and whether they can be applied the same way across all domains. Knowledge is not the discussion point here. Hence, Option B is not the answer.
Option C: The example about remembering strings of digits and letters is not discussed to compare cognitive skills and complex knowledge. We are only looking at how cognitive skills themselves cannot be translatable across arenas. Numbers of digits and letters are different. Cognitive skills in digits don't guarantee the same with letters. Option C is not the answer.
Option D: The example shows that cognitive skills are non-translatable across domains. This has been established in research, and that finding has been replicated. However, the translatability of cognitive skills to a certain domain being replicated in other domains has not been spoken about at all. What's replicated is the psychological research, according to the passage, not translatability. Option D is not the answer. Choice (A)

6. Option A: It can be understood that the claim made by the study was made basis this comparison. If this comparison was not made, the claim couldn't have been made in the first place. Unless success of the individuals can be compared reliably, the study cannot really establish that such generic skills are of not much use. Hence, Option A is not the answer.
Option B: Unless individuals can be trained in generic cognitive skills without domain-specific context, we cannot really conclude whether such a training is useful or not, and whether it

contributes to an individual's success or not. This adds to the reliability of the study, if true. Hence, Option B is not the answer.
Option C: If true, it helps us understand whether the study is really valid or not. If it is possible to spot any link, if it exists, we can tell for sure whether decontextualised skills help directly ***or indirectly later.*** If it is established that such decontextualised learning doesn't help, it strengthens the claims made by the study, since the study is about the success achieved by participants after acquiring certain skills Option C is not the answer.
Option D: This is a new theme, that has got nothing to do with the study, since the study is about those kinds of training which are devoid of any domain specific content. What happens in domain specific training is not related to the study. Hence, Option D is the answer. Choice (D)

7. Option A: This option tends to agree with the statement that as the stakes are higher, at professional levels, as people go higher up in the skill chain, the kind of knowledge needed or expertise needed is also highly specific and not something that is generic, and such skillsets can be taught. Hence, Option A is not the answer.
Option B: Elite athletes (high levels of skill) are given specific mandates – this supports the argument that the expertise needed at high skill levels is also very specific and not everyone can just pick them up and perform well. Hence, Option B doesn't disagree with the given statement. Option B is not the answer.
Option C: Here, because there is a crunch for resources, individuals are making do with whatever they have. They are getting more skilful at what they do because of the time they spend at it. So, it supports the thought process that one can gain skills and experience and expertise across an array of domains and not just limit themselves to one niche area. This possibility weakens the argument presented in the sentence. Hence, Option C is the answer.
Option D: This option proves that an individual's skill level and expertise can be picked up faster based on generic qualities/talents like critical thinking and decision-making abilities. However, this statement doesn't really negate the idea that higher the skill level, more specific the expertise will become. This statement can still hold in case of individuals who are quick learners. Hence, Option D is not the answer. Choice (C)

8. Option A: While this may be true, the passage isn't about this. It is about how generic thinking/critical thinking skills are not as effective as learning domain specific knowledge. The author talks about broadening a student's subject knowledge for niche skills. Option A is not the answer.
Option B: This is the main argument of the passage: that generic thinking skills is ineffective and is over-hyped. What the author advocates is subject intensive knowledge which is important to get really good in one particular domain. The author promotes learning specific skills deeper rather than believing that a generic set of critical thinking skills will work wonders. Option B is the answer.
Option C: Decontextualised practice of cognitive skills is over-hyped according to the author. Nevertheless, the passage doesn't say anywhere that it does more harm than good. The negative consequences of such a practice have not been mentioned. It has only been mentioned that they are useless. Option C is not the answer.
Option D: The author argues against the belief that a generalised set of problem-solving approaches will get the job done. But the author's argument is about how students can improve their domain specific knowledge, and not just critical thinking. The last sentence of the passage clearly proves the author is more keen about the subject-specific knowledge of students than just about their critical thinking skills. That has not been mentioned. Option D is not the answer. Choice (B)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 519

9. Option A: While the Leviathans restore order by compromising democracy and rights, we cannot say one is a replacement for the other. So, it is incorrect to call order and freedom as two choices, with Leviathans preferring one over the other. Similarly, the reverse is not true either, that the Levellers pick freedom over order. It is not that only one of the two (order or freedom) can exist. Option A is not the answer.
Option B: This is apt as Leviathans prefer to keep/maintain/achieve order through a ***reduced amount of*** democracy and rights. The latter, Levellers, advocate for equality and freedom and other rights. Hence, Option B is the answer.
Option C: The passage doesn't talk about Leviathan-view always providing political stability and Leveller-view always offering economic stability. It merely notes that economic instability could weaken the Leviathan hold. Hence, Option C is not the answer.
Option D: While Leviathans are supported by the Grandees, the Levellers advocate for the general public and their rights. So, it is the other way around. Option D is easy to eliminate. Choice (B)

10. Option A: The author has separate advice for Leviathan and Leveller systems. The author doesn't really exhort China (Leviathan system) to move to the other end of the spectrum, that of the Leveller code, which supports open, fraternal societies. Option A is not the answer.
Option B: The author mentions that the roles and opinions of women and the gay community are pivotal. This cannot be translated to infer that the author exhorts that they be given 'greater' rights. Option B is not the answer.

Option C: This is the advice the author gives to Leveller societies, the exact opposite of what China stands for. Option C is not the answer.
Option D: It can be understood that China has a Leviathan-view political system where rights are compromised in exchange for order. However, the economic momentum and evolution of its society could challenge its Leviathan system, according to the author. So, the author's advice would be to keep the unemployment low, in the absence of which the Leviathan contract (order with reduced democracy and right) could be broken. Option D is the answer. Choice (D)

11. Option A: If this is true, it strengthens the author's opinion (and not weakens) that the challenge for Leveller-systems is to maintain their open societies in the face of threat of volatility – economic and political. Option A is not the answer.
Option B: Even if it is true, it doesn't affect the nature of the author's opinion that such a society faces the challenge of maintaining its democratic and open nature. The threat comes from economic and political volatility. So, if they are tightly coupled, it only serves to strengthen what the author said. Option B is not the answer.
Option C: The author's concern was that Leveller-view societies should guard their open and fraternal nature in the face of economic and political volatility. So, if this option statement is true, it disproves the author's theory that it will be a challenge for these open societies to continue remaining so. Option C is the answer.
Option D: The author's advice to Leveller-type societies is to maintain their values in the face of political and economic volatility. That doesn't necessarily depend on whether these two are tightly or loosely coupled. Option D is not the answer. Choice (C)

12. Option A: The para doesn't mention anywhere that China oscillates between Leviathan and Leveller groups (the tension increases as the economic momentum slows, the passage says), although it is known that Russia has a complete Leviathan-type approach. Option A is not the answer.
Option B: While the first part is true, the passage doesn't mention that the economic momentum swings power from one group to another. It simply says that as the momentum slows, the tension between the two groups might grow. Option B is not the answer.
Option C: This is the true difference. As it mentions, Russia has a Leviathan-type approach. However, China is different. It has a Leviathan-orientation, but economic momentum fuels tensions between its two groups. Option C is the answer.
Option D: While the first part of the option is true, the second part, that the order in China depends on its economic momentum is not entirely true. The author mentions that slowing economic momentum will only increase the tussle between the two groups. We are not sure about the order (the kind that Russia has established at the cost of reduced democracy and rights). Option D is not the answer. Choice (C)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 550

13. Option A: Cubism was possible, as we understand from the second para of the passage, because of '*European artists experimenting with abstraction, shattering conventional notions*.' So, we can infer that avant-garde artists (the trendsetters, like Picasso and Braque) rebelling, not necessarily intentionally, against established conventions contributed to the rise of Cubism, and thus, Modern Art. Hence, Option A is not the answer.
Option B: The rapid developments in photography alleviated painting of its need to faithfully depict the natural world. In other words, the responsibility of depicting what was visible to the eye in the natural world was now taken over by photography, which allowed painters to go more abstract, thus paving the way for Cubism. Hence, Option B is not the answer.
Option C: It is not like the aesthetics of Asian and African arts wasn't understood earlier. (We just cannot be sure). Aesthetic revisiting of those arts means that they were seen under a new light. (With greater respect, probably). So, this option hasn't been mentioned as a contributing factor (the difference between discovery and rediscovery has to be appreciated here). Option C is the answer.
Option D: A change of stance did take place towards hitherto ignored styles. These were considered primitive arts, primitive meaning not so sophisticated, or modern. The aesthetic revisiting of these arts inspired Cubists. Hence, Option D is not the answer, as it has been mentioned as a contributing factor. Choice (C)

14. Option A: Painting was limited to off-whites, brown, greys and black. So, it cannot really be said that painting was completely freed of colour. Hence, Option A is easy to eliminate.
Option B: Painting was unchained (freed) from the image one perceives with the eye, according to the passage. It was more about what one perceives with the brain (or the way image was formed on the brain). So, it is not true that painting was unchained or freed from the image ***beyond*** the eye. It was connected to what was beyond the eye. Option B is not the answer.
Option C: They eschewed beauty (threw it out/rejected) as well as idealised forms. The para doesn't connect idealised forms with beauty. They are two different things. So, 'idealised forms of beauty' is a misconstrued phrase. Option C is not the answer.
Option D: This is true, as painting was approached in an investigative way, to appeal to the viewer's

intellect rather than to the viewer's emotion. Painting's objective changed under the Cubists, and its purpose was to appeal to a viewer's sense of reason/rationality. Hence, it was called the 'emancipation of painting', unchained as it was from the eye and the conventions – of idealised forms and beauty.' Hence, Option D is the answer. Choice (D)

15. Option A: While this painting is credited to be the foundational work of Cubism, which was a step away from the known conventions, we cannot be sure if this painting represented a 'drastic shift' from 'all the conventions' of pictorial representation. That will be an exaggeration. Also, such a thing wasn't mentioned as Matisse's reason for using the remark. Hence, Option A is not the answer.
Option B: This was the reason Matisse called it the death of painting, because it undid the progress achieved, the drastic step forward taken, by one-point perspective. So, painting as they had known it had changed. Option B is the answer.
Option C: While multiple perspectives were unprecedented, we cannot infer that that triggered Matisse's remark. Also, the option itself is rather circumlocutory and ambiguous in that it says that the 'perspectives' were never imagined before. That's not true. More than one perspective wasn't used for paintings before, and that is completely different from what the option is saying. Option C is not the answer.
Option D: We are not sure if the one-point perspective can be called the 'foundational work' in painting. Even if so, we are not sure if this painting did 'irreversible damage'. Hence, Option D is not the answer. Choice (B)

16. Option A: This is true, as art's new role in Modern society was to inculcate debate, 'the more heatedly the better', according to the last para. Hence, the author will approve of the statement that the controversy surrounding Cubism was what art was supposed to do. Option A is the answer.
Option B: The connection between Cubism's shift from established norms and the controversy has not been made in the para. It has been mentioned that Cubism raised several debates, but the inevitability of the debates hasn't been attributed to Cubism's shattering of rules. Option B is not the answer.
Option C: The author didn't mention anything about purists in the last para. Also, Cubism was the revolt of the intellect. Cubism's revolt against purism while alluded to earlier in the passage hasn't been explicitly mentioned in the last para of the passage.
Option D: Cubism doesn't instigate intellect; it appeals to it. And Cubism's appeal to the intellect doesn't necessarily raise controversy. Such a thing hasn't been mentioned or discussed anywhere in the passage. Option D is not the answer. Choice (A)

Solutions for questions 17 and 18:

17. Option A: The para comprises of two parts. First, it mentions that democracy is justified if its vital truth is upheld, this vital truth being that human beings can be given equal access to educational opportunity. Secondly, this vital truth involves the individual's right to any source of learning that prepares them for office and power. This option mentions both parts of the para. Hence, Option A is the answer.
Option B: Democracy demands for every individual the right to educational opportunity, this much is not wrong. However, democracy doesn't demand the access to any office that tests humans' fitness for power. It wants the right for humans to any avenue that nourishes them to compete for an office. Hence, Option B is not the answer.
Option C: Democracy rests on the vital truth/assumption that the access to education and opportunity available to humans can be made equal. However, this assumption cannot be extended to include the thought that when the access is made equal humans will become equal. That is an extrapolation not mentioned in the para. Option C is not the answer.
Option D: The para doesn't argue that only humans who are not created equal must get access to education and opportunity equally. It also doesn't argue that all humans are created unequally. Hence, this option doesn't represent the essence of the para. Option D is not the answer. Choice (A)

18. Option A: The para doesn't say that people and environments determine the profitability of a company in a knowledge society. It says that the company's ability to take advantage of the ideas (that people and environments are important) influences the company's profitability. The company has to learn to benefit from people and environment if it has to be profitable. Option A is not the answer.
Option B: While this option is partially true, it doesn't mention that a knowledge society forces a company into such a realisation, and a company's ability to understand and implement these realisations determines its ability to be profitable. Option B is not the answer.
Option C: This option indicates all the ideas in the para: that new technology is an indicator of a knowledge society, a knowledge society demands that companies focus on people as assets and also on environments in which they grow. And this focus is important for their profitability. This option mentions all these ideas. Hence, Option C is the answer.
Option D: The para doesn't say that a knowledge society 'relies' on new technologies, neither does it say that new technologies help a company focus on its environment and employees better. The technology is not the central idea of this para. Option D is not the answer. Choice (C)

Solution for question 19:

19. Sentence 1 is a generic sentence that draws a parallel between a work of art and sunsets: neither have a correct way of interpreting them, since neither have a particular message.
Sentence 2 points to something as 'it' which involves making up rules as we go along. However, sentence 2 makes the connection between rules and 'organised sound'.
Sentence 3 explains how we should look at great art – personally and differently and starts with a positive connector, 'In fact.'
Sentence 4 explains the similarity between listening to a piece of music and writing one.
We can understand that 42 is a block because 2 needs introduction to something that produces sound before we can use 'it'. However, 1 and 3 don't talk about music in general. Both 1 and 3 propagate the same idea, but 1 precedes 3 because of the size of the idea as well as because 3 starts with 'in fact'. 1 talks about what is 'not' art, comparing it with sunsets. Something that it is not, is usually followed by something else that it is. (For example, cinema is not just entertainment, sometimes it is education.) The negative comes first because it doesn't make sense once we already define what it is. Hence, 1342 is the answer.
Option A has the right order. Hence, it is the answer.
Option B places the 42 block in front of the 12 block. It means we are going from something specific, in this case, music, to something very broad, about art, without any connectors. That won't be a logical flow of ideas. Hence, Option B is not the answer.
Option C is incorrect, because while 1, 2 and 3 can be force-fitted to offer a narrative, 4 at the end doesn't make any concluding sense, as it is abrupt and without context. It doesn't offer a coherent end to the para. Option C is not the answer.
Option D splits the discussion of art in 1 and 3 by placing the 42 block in between. So, while 1, 4 and 2 is still a permissible flow of ideas, we cannot go back to 3 again. Hence Option D is not the answer. Choice (A)

Solution for question 20:

20. Option A: This option seems to indicate that organisations and movements are different entities and that the former is built from a place of solidarity, and the latter breeds conflict. However, the author is not differentiating between the two, and extends the idea of organisations to movements. Hence, Option A is not the answer.
Option B: This option reflects the two key ideas of the para: that there will always be some disagreement and conflict in a movement, especially one which is diverse. However, trying to avoid conflict will only make it harder to build movements. Hence, Option B is the answer.
Option C: Conflict doesn't help the diversity of the movement. Rather, a movement that is diverse will witness conflict and disagreement. Also, saying that not all movements have people who agree with each other indicates some movements do have complete agreement, which is also not an idea that reflects the essence of the para. Option C is not the answer.
Option D: This option seems to indicate that only those organisations where organisers fail to understand each other's beliefs witness conflict. However, the para says that conflict and disagreement are part and parcel of a diverse movement. And the movement cannot be built by trying to remove disagreements. This has not been mentioned in the option. Option D is not the answer. Choice (B)

Solutions for questions 21 and 22:

21. Sentence 1 is an independent sentence about the Soviet political lexicon, but it definitely needs some context about when the concept was modified.
Sentence 2 explains the priorities of new Soviet thinking: universal human values and resolution of global problems. This is also an independent sentence, but it needs an answer to 'when'.
Sentence 3 is also an independent sentence that talks about global interdependence and international cooperation being emphasized (where and when).
Sentence 4 gives us the context for the previous three sentences – decline of the Cold War and onset of perestroika.
4 obviously begins the para because it is the only sentence that provides context, the other three sentences more or less explaining the process, goals and priorities. While 1, 2 and 3 all talk about the changes in the policy, it should be noted that 1 explains what was removed and 2 explains what was added. So, 1 follows 4 as it talks about modification, which should precede what it was replaced by (interdependence and cooperation). Sentence 3 comes next because it talks about what was accepted in the place of what was modified. So, 1 and 3 are both about modification and emphasizing whereas the last sentence, 2, is about the essence. 2 comes in the end because it concludes by explaining what the essence of the Soviet thinking was. This makes sense only after we know what the policy was, and what the changes were. Hence, 4132 is the right order.
Option A can be eliminated because there is no context to when 'it was stated'. So, 2 cannot begin the para. Option A is not the answer.
Option B has the right order. Hence, it is the answer.
Option C begins with 2 which is incorrect. Also, 1 won't make sense after 3 because 1 introduces 'Soviet lexicon.' Hence, Option C is not the answer.
Option D: 2 talks about the new Soviet thinking, so it makes sense in the end rather than before other sentences which are still talking about modifications in the older thinking. Hence, Option D is not the answer. Choice (B)

22. Sentence 1 starts with a contrast marker 'In contrast' followed by the violent measures taken by French revolutionaries.
Sentence 2 is an independent sentence that talks about how the American Revolution didn't bring much change.
Sentence 3 talks about what America did after the war, which means 3 chronologically appears after 2, which mentions the American revolution.
Sentence 4 refers to 'it' that was opposed by colonists.
It can be understood that 1 is in contrast to all the other statements because 1 talks about massive changes after a revolution (French), whereas the other three are talking about American revolution and how the change was not much. But, 4 comes before 3 because 4 talks about 'the outset' beginning, whereas 3 talks about 'after the war', the consequences, and later effects. So, the answer is 2431.
Option A can be eliminated because 41 cannot be a block. The contrast in 1 is with respect to American revolution as a whole and has nothing to do with indifference of colonists. Option A is not the answer.
Option B can be eliminated because 1 cannot be followed by 2, because 1 talks about French Revolution, and 2 comes back to American Revolution without any contrast or connectors making the first half of 2 context-less. Hence, Option B is not the answer.
Option C has 3 and 4 together which makes it seem like colonists opposed rebuilding that doesn't make much sense. Hence, Option C is not the answer.
Option D has the right order. Option D is the answer. Choice (D)

Solutions for questions 23 and 24:

23. The bold statement mentions that 'a sizable cohort' ***also*** reinforces something. This is a benefit that should follow a sentence that offers benefits of a sizable cohort. So, it cannot be in the first blank which will make it the first sentence of the para. Similarly, the sentence after Blank 1 doesn't mention any benefits, just that the cohort experience is critical. Hence, it cannot be Blank 2 either where this sentence will fit.
By the same logic, the sentence before the third blank doesn't mention any benefits either, just that new employees will feel better basis certain decisions taken by the management after they are hired. Also, the subsequent sentence starts with 'And' and continues the tone of what makes the new hires without four-year degrees more comfortable. So, Blank 3 cannot be the answer.
The sentence before the fourth blank (which starts with 'And') mentions how having a community bolsters confidence. This is one of the benefits that fits with other advantages of a sizable cohort. This justifies the use of the 'also' in the bold sentence. Hence, Blank 4 is the answer.
The fifth blank also fits with the tone of the bold statement. But it must be noted that the sentence preceding the fifth blank starts with 'one or two people', which makes sense only if the preceding sentence talks about 'a sizable cohort', of which one or two people may feel left out. Otherwise, it won't make sense. So, Blank 4 is a better position. The fifth blank is a close option but not the answer. Ans: (4)

24. It is not the first blank because the bold statement says 'these fields,' which means it must be preceded by a statement that offers some fields. So, it cannot be the first sentence of the para.
It is not blank 2 either because the statement following blank 2 says 'their national identity' pointing to 'Egyptians' in the statement before blank 2. They fit. The bold statement talks about Egypt and not Egyptians.
It cannot be blank 3 because the statement before the blank talks about writers, poets and artists but not any fields in particular. Also, the statement following the blank continues the tone, whereas the bold statement is a much more rounded idea about Egypt as a whole. It cannot be Blank 4 either because, the statement following the fourth blank refers to 'this tolerant attitude' which points to the sentence before blank 4 that talks about religious scholars sanctioning art in schools.
The bold statement fits in the fifth blank because the previous sentence talks about fields like publishing, photography and film industry. It also talks about photography in particular which makes sense in this context. Hence, Option D is the answer. Ans: (5)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	1, 2, 8, 10, 14, 19, 23
Difficult	3, 4, 5, 6, 7, 9, 11, 12, 13, 15, 16, 17, 18, 22, 24
Very Difficult	20, 21

SECTION – II

Solutions for questions 1 to 5:

1. The following table provides the values in the graphs and the totals:

Day	A	B	C	D	E	TOTAL
Monday	100	40	30	50	40	260
Tuesday	80	60	50	40	20	250
Wednesday	70	80	60	30	20	260
Thursday	40	30	60	80	60	270
Friday	30	100	80	40	90	340
Saturday	70	30	90	40	70	300
Sunday	50	70	40	60	40	230
TOTAL	440	410	420	370	360	2000

The maximum total number of persons who watched any movie during the week = 440. Choice (C)

2. Ravi watched five movies on five consecutive days starting from Monday. To maximize the number of Ravi's friends, we can assume that Ravi watched the movies on the days in which the maximum number of persons watched the movie.
The maximum number of persons who watched D on any day in the first five days is 80 (on Thursday).
We can try to see if the number of persons who watched the other movies on other days can also be at least 80.
This is possible if Ravi and his friends watched C on Friday. However, for E, the number cannot be 80 or 90 since Friday and Thursday both cannot be the days on which they could have watched E.
The same problem occurs if we take 70 as well.
The next best option is 60 (since on 3 days 60 persons watched C).
This is possible if they watched A, B, C, D and E on Monday, Tuesday, Wednesday, Thursday and Friday, respectively.
Hence, the maximum number of friends that could have watched the movies along with Ravi is 59 (not counting Ravi). Ans: (59)

3. Ravi must have watched five movies from Wednesday. To maximize the number of Ravi's friends, we can assume that Ravi watched the movies on the days in which the maximum number of persons watched the movie.
The maximum number of persons who watched D on any day in the last five days is 80 (on Thursday).
Ravi and his friends could have watched C, D and E on Saturday, Thursday and Friday, respectively. However, 80 persons could not have watched A on either Wednesday or Sunday. Hence, this is not possible.
If 70 persons watched D on Thursday, they could have watched A, B, C and E on Wednesday, Sunday, Saturday and Friday.
Hence, the maximum number of friends who could have watched the movies on these days = 70 – 1 = 69. Ans: (69)

4. The five movies were watched for a total of 2000 times. We need to maximize the number of persons who watched each movie more than once. Since it is given that every person watched 5 movies, we need to maximize the number of persons who watched 6 times (four movies once and one movie twice).
The maximum number of persons possible, who can watch 6 times = 2000/6 = 333.33
Assume that there are 333 persons. Each person would have watched all the five movies exactly once.
Since A was watched for a total of 440 times, we can assume that a maximum of 440 – 333 = 107 persons watched A more than once. These 107 persons are a set of persons among the 333 persons.
Hence, 333 – 107 = 226 persons watched A once and 107 persons watched A more than once.
Similarly, for B, the number of persons who could have watched B more than once = 410 – 333 = 77. Hence, a maximum of 77 persons could have watched B more than once.
For C, a maximum of 420 – 333 = 87 persons could have watched more than once.
For D, a maximum of 370 – 333 = 37 persons could have watched more than once.
For E, a maximum of 360 – 333 = 27 persons could have watched more than once.
To maximize the total number of persons who watched any movie more than once, we can take all the above persons to be distinct.
The total number of persons = 107 + 77 + 87 + 37 + 27 = 335.
However, since there are only 333 persons, the maximum number of persons who could have watched any movie more than once = 333. Choice (A)

5. By observing the values from the graph, we can see that the number of persons who watched B was greater than the number of persons who watched A on 3 days (Wednesday, Friday and Sunday). On each of these three days, the number of persons who watched B was greater than that of E. Hence, the given condition is satisfied for none of the days. Choice (A)

Solutions for questions 6 to 10:

Given that, if 6 students were distributed among the classes 7, 8, 9 and 10, these four classes will have an equal number of students.
Since the number of students in these four classes are distinct, the number of students that must be added to each class to get an equal number of students must also be distinct. The only possibility to have four distinct numbers adding up to 6 is 0, 1, 2 and 3.
If the number of students in one of the four classes is a, then the number of students in the other three classes must be a – 1, a – 2 and a – 3.
Now, if 7 students were distributed among the classes 6, 8, 9 and 10, these four classes will have an equal number of students.
If there are a students in class 7, then to have an equal number of students in classes 8, 9 and 10, we need to add 1, 2 and 3 students OR 0, 1, 2 students to these classes (since the number of students in these three classes will be a – 1, a – 2 and a – 3). In the first case, the number of students in Class 6 must be a – 1 (since we can add only one more student). However, this is not possible as this is already the number of students in one of the other classes.
In the second case, the number of students in Class 6 must be a – 5. This is possible.
If the number of students in Class 7 is a – 1, the number of students in classes 8, 9 and 10 will be a, a – 2 and a – 3, in any order. For the number of students in these three classes to be equal we need to

add 0, 2 and 3 students. This implies that the number of students in Class 6 will be a – 2 (as we can add only 2 more students to this class). However, this is not possible.
If the number of students in Class 7 is a – 2, the number of students in classes 8, 9 and 10 will be a, a – 1 and a – 3, in any order. For the number of students in these three classes to be equal we need to add 0, 1 and 3 students OR 1, 2 and 4 students. In the first case, the number of students in Class 6 will be a – 3 (as we can add only 2 more students to this class). However, this is not possible. In the second case, the number of students in Class 6 will be a + 1 (as we cannot add any more students to this Class). This case is possible.
If the number of students in Class 7 is a – 3, the number of students in classes 8, 9 and 10 will be a, a – 1 and a – 2, in any order. For the number of students in these three classes to be equal we need to add 0, 1 and 2 students. This implies that the number of students in Class 6 will be a – 4. This case is possible.
Hence, there are three cases possible: Class 6 has a – 5 students, Class 7 has a students OR Class 6 has a + 1 students, and Class 7 has a – 2 students OR Class 6 has a – 4 students and Class 7 has a – 3 students.
If we add 8 students to classes 6, 7, 9 and 10, there must be an equal number of students.
In the first case, we have to add 5 students to Class 6 to make it equal to that in Class 7. The remaining 2 students must be added to Class 9 and Class 10. So Class 9 and Class 10 can have a – 1 and a – 2 students in any order. Hence, Class 8 must have a – 3 students.
In the second case, we have to add 1, 3 and 4 students to 7, 9 and 10. To Class 6 we will not add any students. Hence, Class 8 must have a – 1 students.
In the third case, we have to add one more student to either of Class 9 or Class 10. Hence, the number of students in Class 9 and Class 10 must be a – 1 and a in any order. The number of students in Class 8 must be a – 2.
If we add 9 students to classes 6, 7, 8 and 10, there must be an equal number of students.
This implies that in the first case, the number of students in Class 9 must be a – 2 and that in Class 10 must be a – 1.
In the second case, the number of students in Class 9 will be a and that in Class 10 will be a – 3.
In the third case, the number of students in Class 9 must be a – 1 and that in Class 10 must be a.
The three possible cases for the number of students are presented in the table below:

Class	Case 1	Case 2	Case 3
Class 6	a – 5	a + 1	a – 4
Class 7	a	a – 2	a – 3
Class 8	a – 3	a – 1	a – 2
Class 9	a – 2	a	a – 1
Class 10	a – 1	a – 3	a

Since we know that there must be at least 6, 7, 8, 9 and 10 students in Class 6, 7, 8, 9 and 10. By observation, we can see that the minimum value of a in case 1 is 11, in case 2 is 13 and in case 3 is 10.

6. Statement I is true only for Case 2. In this case, the highest number of students are in Class 9.
Statement II is true for Case 1 and Case 3. In these cases, the highest number of students can be in Class 7 or 10.
Hence, the question can be answered using Statement I alone but not using Statement II alone. Ans: (2)

7. From Statement I, in Case 1, a – 1 is a multiple of 10.
From Statement II, in Case 1, a – 5 is a multiple of 8.
The minimum possible value of a is 21.
For a = 21 or any value greater than 21, the number of students in all the classes will be greater than 15.
From statement I, in Case 2, a – 3 is a multiple of 10.
From statement II, in Case 2, a + 1 is a multiple of 8.
Hence, a can be a minimum of 23. The number of students in all classes will be greater than 15.
From Statement I, in Case 3, we can say that a is a multiple of 10.
From Statement II, in Case 3, a – 4 is a multiple of 8.
The minimum possible value of a is 20.
For a = 20 or any value greater than 20, the number of students in all the classes will be greater than 15.
Hence, the question can be answered using both the statements together. Ans: (4)

8. Statement I:
From this statement, we can eliminate Case 2, but we will not get any additional information.
Statement II:
Case 1: a – 1 = 2a – 10 – 8 ⇒ a = 17.
Number of students in Class 9 = a – 2 = 15.
Case 2: The condition is not applicable for Case 2.
Case 3: 2a – 8 – 8 = a ⇒ a = 16.
Number of students in Class 9 = a – 1 = 15.
Hence, the question can be answered using the second statement alone but not by using the first statement alone. Ans: (3)

9. In each of the three cases, the number of students in Class 9 is greater than that in Class 8. Hence, the question can be answered without any additional information. Ans: (6)

10. In all the three cases, the difference between the number of students in Class 8 and the number of students in Class 6 = 2.
This question can be answered without any additional information. Ans: (6)

Solutions for questions 11 to 15:

Given that the rank of A increased in Test 2 as compared to his rank in Test 1. Hence, in Test 2, A's rank can be 2/3/4/5. In Test 1, his rank can be 1/2/3/4. In Test 3, his rank remained unchanged. Hence, in Test 3, his rank can be 2/3/4/5.
In Test 4, his rank decreased. Hence, his rank in Test 4 can be 1/2/3/4.
In Test 5, his rank decreased again. However, if his rank in Test 4 was 1, it cannot decrease further. Hence, his rank in Test 4 cannot be 1. Also, in Test 3, his rank cannot be 2 because it cannot decrease further in Test 4. Hence, A's rank in Test 2 and Test 3 must be 3/4/5.
In Test 4, it must be 2/3/4. In Test 5, it must be 1/2/3.
B's rank decreased in Test 2, Test 4 and Test 5 and remained the same in Test 3.
B's rank in Test 2 (and Test 3) must be at least 3 (only then it can decrease in Test 4 as well as Test 5).
B's rank in Test 2 can be 3/4 (it cannot be 5 since it decreased in Test 2). In Test 3 also, B's rank can be 3/4.
In Test 4, B's rank can be 2/3. In Test 5, B's rank can be 1/2.
C's rank decreased in Test 2 and Test 3. Hence, C's rank in Test 2 can be 2/3/4. In Test 1, it must be 3/4/5.
In Test 3, C's rank can be 1/2/3.
C's rank increased in Test 4 and Test 5. Hence, in Test 4, C's rank can be 2/3/4. In Test 5, C's rank can be 3/4/5.
D's rank decreased in Test 3. Hence, D's rank in Test 3 can be 1/2/3/4. In Test 1 and Test 2, D's rank can be 2/3/4/5.
In Test 4, D's rank can be 2/3/4/5. In Test 5, D's rank can be 1/2/3/4.
E's rank in Test 2 can be 1/2/3/4. In Test 3, it must be 2/3/4/5. In Test 4, it must be 1/2/3/4. In Test 5 also, it must be 1/2/3/4.
The following table provides the possible ranks of the students in the tests:

Student	Test 1	Test 2	Test 3	Test 4	Test 5
A	1/2/3/4	3/4/5	3/4/5	2/3/4	1/2/3
B	4/5	3/4	3/4	2/3	1/2
C	3/4/5	2/3/4	1/2/3	2/3/4	3/4/5
D	2/3/4/5	2/3/4/5	1/2/3/4	2/3/4/5	1/2/3/4
E	2/3/4/5	1/2/3/4	2/3/4/5	1/2/3/4	1/2/3/4

In Test 4, E is the only person who can have rank 1. Hence, E's rank in Test 4 must be 1. Since E's rank remained unchanged in Test 5, it must also be 1 in Test 5.
In Test 5, B's rank cannot be 1. Hence, B's rank in Test 5 must be 2. In Test 4, B's rank must be 3 (since the rank of B decreased in Test 5, this is the only possibility). In Test 3, B's rank must be 4. In Test 2, B's rank must be 4. In Test 1, it must be 5.
In Test 5, A's rank cannot be 1 or 2. Hence, A's rank must be 3. A's rank in Test 4 must be 4 (since A's rank decreased in Test 5). In Test 3, A's rank must be 5. In Test 2, A's rank must be 5. In Test 1, it can be 1/2/3/4.
In Test 5, only C can have a rank of 5. Hence, C's rank in Test 5 must be 5. D's rank in Test 5 must be 4.
In Test 4, C's rank can only be 2 (since A and B are ranked 4 and 3 respectively). D's rank must be 5 in Test 4.
C's rank in Test 4 increased as compared to his rank in Test 3. Hence, C's rank must be 1 in Test 3. Rank of D and E in Test 3 can be 2/3 in any order.
In Test 2, the only person who can have a rank of 1 is E. Hence, E's rank in Test 2 is 1. The rank of C and D in Test 2 can be 2/3 in any order. But D's rank decreased in Test 3 as compared to his rank in Test 2. Hence, D's rank in Test 2 must be 3 and his rank in Test 3 must be 2. E's rank in Test 3 must be 3. C's rank in Test 2 must be 2.
D's rank in Test 1 is the same as his rank in Test 2. Hence, D's rank in Test 1 is 3. C's rank in Test 1 must be 4 (since B and D are ranked 5 and 3 respectively). In Test 1, E's rank (since E's rank decreased from Test 1 to Test 2) must be 2 and A's rank must be 1.
The following table provides the ranks of each of the five students in each of the five tests:

Student	Test 1	Test 2	Test 3	Test 4	Test 5
A	1	5	5	4	3
B	5	4	4	3	2
C	4	2	1	2	5
D	3	3	2	5	4
E	2	1	3	1	1

11. B received the fifth rank in Test 1. Choice (A)

12. D's rank in Test 2 is 3. Choice (B)

13. All the remaining 4 students received a higher rank than E in at least four of the five tests. Choice (D)

14. The maximum sum of the ranks (for A and B) is 18. Choice (B)

15. C received the second rank in Test 4. Choice (C)

Solutions for questions 16 to 20:

Given that there are 4 persons between any two persons who have 4 chocolates each.
Let the arrangement of these two persons be 4 _ _ _ _ 4. Apart from these six persons, there must be two more persons on any of the two sides.
The person in the second place in the above line cannot have 3 chocolates. This is because if he has 3 chocolates, then there must be another person with three chocolates four places to his left, which is not possible as there cannot be more than 8 persons in the line, OR four places to his right, which is not possible as this person has 4 chocolates.

Consider that the person in the second place has 1 chocolate. Then the following cases are possible:
1 4 1 _ _ _ 4 OR 4 1 _ 1 _ 4
In the first case, the person to the right of the person with 1 chocolate cannot be 2 (since none of the places three places away are free). Hence, this person has to be 3. This implies that the two vacant places to the right of this 3 must be 2's. However, that is not possible. Hence, the first case is not possible.
In the second case, the person between the two 1's can be 2. We get 4 1 2 1 _ 4.
The other 2 must be at the extreme left. Hence, we get 2 4 1 2 1 _ 4.
The blank in the middle must be 3. However, the other 3 cannot be fitted in any place in this line. Hence, the person in the second place cannot have 1 chocolate.
Consider that the person in the second place has 2 chocolates. The following cases are possible:
4 2 _ _ 2 4 OR 2 4 2 _ _ _ 4.
In the first case, the two persons in the middle both cannot have 1 chocolate. Hence, they both must have 3 chocolates, which is not possible.
In the second case, the person to the right of the person at the extreme left must have 3 chocolates. Hence, 2 3 4 2 _ _ _ 4.
In the remaining places, there is only one way in which we can fill the numbers: 2 3 4 2 1 3 1 4.
We can get the same sequence in reverse 4 1 3 1 2 4 3 2. But this can be ruled out from (iii).
Hence, the eight persons, A through H, must have 2, 3, 4, 2, 1, 3, 1 and 4 chocolates.
In the case of biscuits, let the persons with 5 biscuits be represented with the following arrangement:
5 _ _ _ _ 5.
The second person cannot have 4 biscuits.
If the second person has 3 biscuits, then the following arrangement is possible: 5 3 _ _ _ 3 5.
In this case, the third person can have 0 biscuits or 4 biscuits. If the third person has 0 biscuits, the 4th person also must have 0 biscuits. The 5th person must have 4 biscuits. In this case, the other person with 4 biscuits must be to the immediate left of the person who has 5 biscuits (the person at the extreme left in the above arrangement.) The arrangement will be 4 5 3 0 0 4 3 5. Then A will have 4 biscuits and H will have 5 biscuits. From (iii), this is not possible.
If the third person has 4 biscuits, then the following arrangement is possible: 5 3 4 _ _ 3 5 4. The other two persons must have 0 biscuits. Hence, one possible arrangement is 5 3 4 0 0 3 5 4.
If the second person has 0 biscuits, then the third person also must have 0 biscuits. The fourth person cannot have 4 biscuits because the other person with 4 biscuits must be standing two places to the right of the person with 5 biscuits (at the right). This increases the number of persons to more than 8. Hence, the fourth person must have 3 biscuits, and there must be a person at the extreme right with 3 biscuits. The arrangement will be 5 0 0 3 _ _ 5 3. However, in this case, the two persons with 4 biscuits must be standing adjacent each other. Hence, this case is also not possible.
Therefore, the only possible case exists for biscuits is 5 3 4 0 0 3 5 4.
We can get the same sequence in reverse 4 5 3 0 0 4 3 5. But this can be ruled out from (iii).
The following table provides the number of chocolates and biscuits with each person:

Person	A	B	C	D	E	F	G	H
Chocolates	2	3	4	2	1	3	1	4
Biscuits	5	3	4	0	0	3	5	4

16. D has 2 chocolates. Choice (B)

17. G has the highest number of biscuits. Choice (D)

18. The number of chocolates is the same as the number of biscuits for 4 persons (B, C, F, H). Choice (C)

19. The maximum number of biscuits with the person standing adjacent to any person with 1 chocolate is 4 (with H, who is standing adjacent to G). Choice (C)

20. E has the minimum sum of the number of chocolates and biscuits, i.e., 1. Choice (A)

Difficulty level wise summary - Section II

Level of Difficulty	Questions
Very Easy	5
Easy	1
Medium	2, 3, 16, 17
Difficult	4, 11, 12, 13, 14, 15, 18, 19, 20
Very Difficult	6, 7, 8, 9, 10

SECTION – III

Solutions for questions 1 to 3:

1. Let the least number be N. 1 is its least divisor. Let its 2nd, 3rd and 4th least divisors be a, b and c respectively.
We consider the following values for the divisors x and the corresponding values of x^3. From a, b, c exactly 1 or all 3 are odd (∵ N is even)

x	x^3
1	1
2	8
3	27
4	64
5	125
6	216

For (a, b, c) = (2, 3, 4), 2N = 100 (i.e., N = 50). But 3 is not a divisor of 50. For (a, b, c) = (2, 3, 6), 2N = 252 (i.e., N is 126) and 1, 2, 3, 6 are the four least distinct divisors of 126. The required number is 126. The sum of the digits is 9. Ans: (9)

2. Volume of the cuboid = $(4^2 \times 6) = 96$
Volume of the cylinder = $\pi r^2 h = \pi\left(2\sqrt{2}\right)^2 (6) = 48\pi$

Unoccupied volume = 48π – 96
The gap between the cylinder and cube $= \left(4\sqrt{2}\right) - \frac{4}{2}$

i.e., ≈ 0.8 inches. Therefore only one sphere can be placed in the gap by considering the diameter.
As the height is 6 inches, a total of $4 \times \left[\frac{6}{0.8}\right]$, i.e.,
4 × 7 = 28 spheres can be placed. Ans: (28)

3. x + y + z + 6 where x > 0, y > 0 and z < 0
Let w = –z. ∴ x + y + w = 6 . (1), where x > 0, y > 0, w > 0.
We need the maximum value of xy^3w^2 $(z^2 = w^2)$
We need to split the y in (1) into 3 parts and the w into 2 parts.
$(1) \Rightarrow x + 3\left(\frac{y}{3}\right) + 2\left(\frac{w}{2}\right) = 6$
The sum of these 6 positive parts $\left(x, \frac{y}{3}, \frac{y}{3}, \frac{y}{3}, \frac{w}{2}, \frac{w}{2}\right)$ is constant. The maximum value of the product occurs when they are all equal (and therefore equal to 1)
i.e., x = 1, $\frac{y}{3} = 1$ and $\frac{w}{2} = 1$
The maximum value of xy^3z^2 is $1(3^3)(2^2) = 108$. Ans: (108)

Solutions for questions 4 and 5:

4. The mean of 23.5, 27.5, 26.5, 25.5 and 24.5 equals 25.5
When x equals the mean of the rest of the values, the standard deviation will be the minimum.
i.e., x = 25.5.
$$S.D. = \sqrt{\frac{(23.5 - 25.5)^2 + \ldots\ldots + (27.5 - 25.5)^2}{6}}$$
$$= \sqrt{\frac{2^2 + 1^2 + 0^2 + 0^2 + 1^2 + 2^2}{6}} = \frac{\sqrt{5}}{\sqrt{3}}$$ Choice (C)

5. $$3 + \cfrac{1}{3 + \cfrac{3}{3 + \cfrac{3}{3 + \cfrac{3}{2}}}} = 3 + \cfrac{1}{3 + \cfrac{3}{3 + \cfrac{3}{\left(\frac{11}{3}\right)}}}$$
$$= 3 + \cfrac{1}{3 + \cfrac{3}{\left(\frac{42}{11}\right)}} = 3 + \cfrac{1}{3 + \cfrac{3}{\left(\frac{159}{42}\right)}}$$
$$= 3 + \cfrac{1}{3 + \frac{126}{159}}$$
$$= 3 + \frac{159}{603} = \frac{1968}{603}$$

Alternative Solution:

This question may also be attempted using the basic on-screen calculator provided. However, given that the answer choices are very close to each other, it would be necessary to identify the correct answer by checking up to the fourth decimal place. Choice (D)

Solution for question 6:

6. Amar runs only 950 m, when Tarun runs for 1000 m.
Now Tarun runs only 960 m,
∴ Amar runs for $\frac{950 \times 960}{1000} = 912$ m.
Hence, Tarun beats Amar by 88 m. Ans: (88)

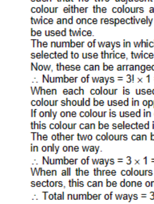
Solutions for questions 7 to 9:

7. Let the number of model A's manufactured be x.
Total time spent on manufacturing Model A's = 4x hours. Total time available for manufacturing model B's = 1600 – 4x hours. Number of model B's which can be manufactured
$= \frac{1600 - 4x}{2} = 800 - 2x$.
Total profit earned on both models
= 1200x + 1000(800 – 2x) = 800000 – 800x
This will be maximum when x = 0.
Hence, total number of Model B's should be 800 – 0 = 800 for maximum profit. Choice (C)

8. Given, (x – m) (x – n) – 4 = 0
To comment about the roots of the above equation, let us re-write the same as (x – m) (x – n) = 4.
This means that (x – m) (x – n) > 0.
Now, the standard results of inequalities say that when (x – m) (x – n) > 0, x must lie outside of the roots(that is, x must be either less than the smaller root or greater than the larger root).
In other words, if the expression (x – m) (x – n) is positive (note that the exact value of the expression – i.e., '4' or any other positive value – is not relevant any more), and m is greater than n, then x must lie in the range (–∞, n) and (m, ∞).
Therefore, when (x – m) (x – n) = 4, the roots will lie in the range (–∞, n) and (m, ∞).
[Also, when (x – m) (x – n) < 0, x must lie inside/between the roots (that is, x must be greater than the smaller root and less than the larger root).]

Alternative solution 1:

Consider the diagram shown below:



Let y = (x – m) (x – n) – 4 ------ (1)
In the above equation, for x = m and x = n, we get y = –4.
Again (x – m) (x – n) – 4 = x^2 – (m + n)x + mn – 4
The discriminant for the quadratic equation y = 0 is $\{-(m + n)\}^2 - 4(1)(mn - 4)$
$= m^2 + n^2 - 2mn + 16 = (m - n)^2 + 16$
Since the discriminant is greater than 0, the roots are real and distinct.
Let the roots be α and β, where α > β.
Now, the co-efficient of x^2 is 1, (i.e., positive). The graph is a parabola that opens upwards as shown in the figure above.
We can see that the roots lie in (–∞, n) and (m, ∞).

Alternative solution 2:

With some basic understanding of graphs, this question can be solved very easily by first imagining the graph of (x – m) (x – n) = 0, which will intersect the x-axis at x = m and x = n, i.e., (m, 0) and (n, 0).
Now, the graph of y = (x – m) (x – n) – 4 = 0 is simply the original graph being pulled downwards by 4 units.
In the process, clearly the new points of intersection of the curve with the x-axis will now move away from each other in opposite directions.
Thus, one of the roots of (x – m) – (x – n) – 4 = 0 will lie between –∞ and n and the other root will lie between m and +∞ (given that m > n). Choice (B)

9. Let the number of people in the group be n.
Total age = n × 2n = $2n^2$
When X leaves, the total age = (n – 1) × 2(n – 1) = $2(n - 1)^2$
X's age = $2n^2 - 2(n - 1)^2$
when Y leaves, the total age = (n – 2) × 2(n – 2) = $2(n - 2)^2$
Y's age = $2(n - 1)^2 - 2(n - 2)^2$
∴ Ratio of ages of X and Y
$\Rightarrow \frac{2n^2 - 2(n-1)^2}{2(n-1)^2 - 2(n-2)^2} = \frac{19}{17}$
$\Rightarrow \frac{n^2 - (n^2 - 2n + 1)}{(n^2 - 2n + 1) - (n^2 - 4n + 4)} = \frac{19}{17}$
$\Rightarrow \frac{2n - 1}{2n - 3} = \frac{19}{17} \Rightarrow n = 10$
When 'Z' leaves, total age = $2(n - 2)^2 - 16$ = $2(8)^2 - 16 = 112$
Average age = $\frac{112}{7} = 16$ Choice (C)

Solution for question 10:

10. Since a, b, c and b, c, d are in continued proportion and b : c = 2 : 3, we can work out the composite ratio as shown in the table below.

a	b	c	d
2	3		
	2	3	
		2	3
4	6	9	
8	12	18	27

Since a, b, c, d are positive integers a + d = 8 + 27 = 35 Ans: (35)

Solutions for questions 11 and 12:

11. For colouring five sectors with three different colours such that no two adjacent sectors are of same colour, two colours are to be used twice and one colour once.
This can be done in 3 ways.
The number of ways of selecting 2 sectors from 5 sectors for painting a colour = $^5C_2$.
But out of these, five pairs are adjacent sectors.
∴ Non adjacent pairs = $^5C_2$ – 5 = 10 – 5 = 5
Out of the remaining three sectors, two sectors are adjacent. Hence the other colour which is used for another pair of non-adjacent sides. There are two such pairs of non-adjacent sides i.e., 2 ways.
In the remaining sector the third colour can be filled in 1 way.
∴ Total number of ways = 3 × 5 × 2 = 30. Choice (B)

12. We can use either 2 colours or 3 colours to paint the sectors.
First, if only two colours are used to paint the sectors, there is only one possible arrangement - alternate sectors will be in the same colour. That is, if the sectors are a, b, c, d, e and f, in that order, then a, c, and e will be in one colour while b, d, and f will be in the other colour. Hence, there is only one possibility for the arrangement. But the colours can be selected in $^3C_2$ ways, i.e., 3 ways.
For colouring six sectors with 3 different colours such that no two adjacent sectors are of same colour either the colours are to be used thrice, twice and once respectively or each colour should be used twice.
The number of ways in which three colours can be selected to use thrice, twice and once is 3!.
Now, these can be arranged only in one way.
∴ Number of ways = 3! × 1 = 6
When each colour is used twice, at least one colour should be used in opposite sectors.
If only one colour is used in opposite sectors, then this colour can be selected in 3 ways.
The other two colours can be filled in four sectors in only one way.
∴ Number of ways = 3 × 1 = 3
When all the three colours are used in opposite sectors, this can be done only in two ways.
∴ Total number of ways = 3 + 6 + 3 + 2 = 14. Ans: (14)

Solutions for questions 13 and 14:

13. Let the product of all the n numbers be p.
Now, let a, b, c be the three numbers chosen by Rajith, Satish and Tarun respectively. And let q be the product of the remaining n – 3 numbers.
Let each of the three students increase the respective number that they chose by the same quantity, x.
And let p_1, p_2 and p_3 be the increased product in the respective case.
Hence, p = abcq, p_1 = (a + x) bcq, p_2 = (b + x) acq
p_3 = (c + x) abq
Given $\frac{xbcq}{abcq} = \frac{x}{a} = \frac{9}{100}$, $\frac{xacq}{abcq} = \frac{x}{b} = \frac{12}{100}$
and $\frac{xabq}{abcq} = \frac{x}{c} = \frac{18}{100}$
⇒ 9a = 12b = 18c
∴ a : b : c = $\frac{1}{9} : \frac{1}{12} : \frac{1}{18}$ = 4 : 3 : 2 Choice (C)

14. Given $S = 1 + \frac{2}{7} + \frac{4}{7^2} + \frac{7}{7^3} + \frac{11}{7^4} + \ldots \rightarrow$ (1)
Dividing by 7,
$\frac{S}{7} = \frac{1}{7} + \frac{2}{7^2} + \frac{4}{7^3} + \frac{7}{7^4} + \ldots \rightarrow$ (2)
(1) – (2) ⇒ $S - \frac{S}{7} = 1 + \frac{1}{7} + \frac{2}{7^2} + \frac{3}{7^3} + \frac{4}{7^4} + \ldots$
$\frac{6S}{7} = 1 + \frac{1}{7} + \frac{2}{7^2} + \frac{3}{7^3} + \frac{4}{7^4} + \ldots \rightarrow$ (3)
Again dividing by 7.
$\frac{1}{7}\left(\frac{6S}{7}\right) = \frac{1}{7} + \frac{1}{7^2} + \frac{2}{7^3} + \frac{3}{7^4} + \ldots \rightarrow$ (4)
(3) – (4) ⇒ $\frac{36}{49}(S) = 1 + \frac{1}{7^2} + \frac{1}{7^3} + \frac{1}{7^4} + \ldots$
This is in G.P. with $a = \frac{1}{7^2}$ and $r = \frac{1}{7}$
$\Rightarrow \frac{36}{49}(S) = 1 + \left(\frac{1}{7^2}\right)\left(\frac{1}{1 - \frac{1}{7}}\right) = 1 + \frac{1}{42} = \frac{43}{42}$
$\Rightarrow S = \frac{43}{42} \times \frac{49}{36} = \frac{301}{216}$ Choice (A)

Solution for question 15:

15. Let the number of ₹1 coins, ₹2 coins and ₹5 coins in the bag be x, y and z respectively.
x + y + z = 150 ——— (1)
Given, value of ₹1 coins is at least 50% of total.
⇒ x ≥ $\frac{1}{2}$ (x + 2y + 5z)
⇒ 150 – y – z ≥ 2y + 5z
⇒ 50 ≥ y + 2z ——— (2)

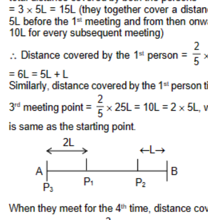
The data is tabulated below:

Denomination	1	2	5
Number	x	127 – x	23
Value	x	254 – 2x	115

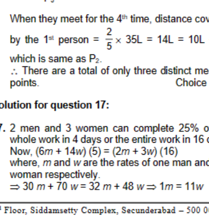
Given z = 23 and from (2), i.e., y + 2z ≤ 50
∴ y ≤ 4, i.e. y = 1, 2, 3 or 4.
If y = 4, x = 123 and the value of the 2 rupee coins is 8/(123 + 8 + 115) or more than 3%.
If y = 1, 2 or 3, this percentage is less than 3%. Ans: (4)

Solution for question 16:

16. Since the point of interest is only the number of meeting points, for the purpose of calculation, interchanging both speeds and directions is equivalent to continuing in the same direction without changing speeds.



Given ratio of speeds is 2 : 3.
Let the length of the track be 5L.
∴ Distance covered by the 1st person form the beginning to the 1st meeting point = $\frac{2}{5} \times 5L = 2L$.
From the starting point to the 2nd meeting point, total distance covered by both the persons = 3 × 5L = 15L (they together cover a distance of 5L before the 1st meeting and from then onwards, 10L for every subsequent meeting)
∴ Distance covered by the 1st person = $\frac{2}{5} \times 15L$ = 6L = 5L + L
Similarly, distance covered by the 1st person till the 3rd meeting point = $\frac{2}{5} \times 25L = 10L = 2 \times 5L$, which is same as the starting point.



When they meet for the 4th time, distance covered by the 1st person = $\frac{2}{5} \times 35L = 14L = 10L + 4L$ which is same as P_2.
∴ There are a total of only three distinct meeting points. Choice (A)

Solution for question 17:

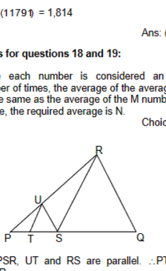
17. 2 men and 3 women can complete 25% of the whole work in 4 days or the entire work in 16 days.
Now, (6m + 14w) (5) = (2m + 3w) (16)
where, m and w are the rates of one man and one woman respectively.
⇒ 30 m + 70 w = 32 m + 48 w ⇒ 1m = 11w

Ratio of the rates of one man and 2 women = 11 : 2
Share of 2 women out of total wages
$= \left(\frac{2}{11 + 2}\right)(11791)$
$= \frac{2}{13}(11791) = 1,814$ Ans: (1814)

Solutions for questions 18 and 19:

18. Since each number is considered an equal number of times, the average of the averages will be the same as the average of the M numbers. Hence, the required average is N. Choice (D)

19.



In ΔPSR, UT and RS are parallel. ∴PT/PS = PU/PR.
In ΔPQR, US and RQ are parallel.
∴ PS/PQ = PU/PR = 2/3 (given)
∴(PT + TS) : SQ = 2 : 3 = 10 : 15
and PT : TS = 2 : 3 = 4 : 6
∴ (PT : TS) : SQ = 4 : 6 : 15 and TS : SQ = 2 : 5 Choice (B)

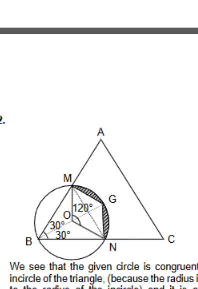
Solutions for question 20:

20. $104^n = [51(2) + 2]^n = M(51) + 2^n$, where M(51) denotes some multiple of 51.
Taking n = 1, 2, 3, 4, 5, 6, 7, 8, we get the possible values of the remainder when 104^n is divided by 51 as 2, 4, 8, 16, 32, 13, 26, 1 respectively.
For n > 8, the pattern repeats. Therefore, the maximum possible value of the remainder is 32. Ans: (32)

Solutions for questions 21 and 22:

21. Let a w pound water-melon cost ₹c.
Given that
c ∝ $\sqrt{w}$ ⇒ c = k$\sqrt{w}$
when w = 16, c = 50
⇒ 50 = k$\sqrt{16}$ ⇒ k = 12.5 and ∴ c = 12.5$\sqrt{w}$
∴ $\sqrt{w} = \frac{37.5}{12.5} = 3 \Rightarrow w = 9$
∴It weighs 9 pounds. Choice (C)

22.



We see that the given circle is congruent to the incircle of the triangle, (because the radius is equal to the radius of the incircle) and it is also the circumcircle of ΔMBN.
If r, a, R are the inradius, side and circumradius of an equilateral triangle,
$r = \frac{a}{2\sqrt{3}}$ and $R = \frac{a}{\sqrt{3}}$
Given, a = $2\sqrt{3}$ ⇒ r = 1
Area of the circle that remains after the triangle is cut off = 2 (Area of minor segment on BN)
$= 2\left[\left(\frac{1}{3}\right)\pi r^2 - \text{Area of } \Delta BON\right]$
Now, area of ΔBON = $\frac{1}{3}$ area of ΔBMN
$= \left(\frac{1}{3}\right)$ [1/4 of Area of ΔABC] = $\left(\frac{1}{3}\right)\left(\frac{1}{4}\right) 3\sqrt{3}r^2$
(from (2) in the earlier solutions)
⇒ Area of ΔBON = $\frac{\sqrt{3}}{4}r^2$
∴ Remaining area of circle =
$2\left[\left(\frac{1}{3}\right)\pi r^2 - \frac{\sqrt{3}}{4}r^2\right] = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$ (∵ r = 1)
$= \frac{4\pi - 3\sqrt{3}}{6} \approx 1.23$ Choice (B)

Difficulty level wise summary - Section III

Level of Difficulty	Questions
Very Easy	6
Easy	5, 10, 17, 18, 21
Medium	2, 4, 7, 8, 13, 15, 19
Difficult	1, 9, 14, 16, 20, 22
Very Difficult	3, 11, 12

Done
Tip: Scroll slowly for the best results